COURSE SYLLABUS

# ENGR 1105: Intro to Robotics & Mechatronics

Fall 2026

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| **Instructor** | Dr. Huan Phan, Assistant Professor of Robotics and Mechatronics Engineering |
| **Email** | hphan206@randolphcollege.edu |
| **Class meetings** | **Mon., Tue., Thu., Fri. • 9:30–10:45 a.m. • Martin Science Building, Room 208** |
| **Course dates** | October 26–December 14, 2026 |
| **Office hours** | By appointment; weekly times posted on the course website |
| **Course site** | `huantianh.github.io/courses/intro-robotics-mechatronics`<br>Announcements, readings, lab instructions, assignments, and schedule updates |

## Course Description

This hands-on course introduces robotics and mechatronics through electronics, Arduino programming, sensors, actuators, and control. Lectures directly prepare students for weekly Arduino laboratories. Students also complete a separate, self-directed mechanical-automaton project and a final controller-based Arduino project. No previous programming or electronics experience is required.

## Course Learning Outcomes

By the end of the course, a successful student will be able to:

1. explain the roles of structures, mechanisms, machines, mechatronic systems, and robots;
2. apply the engineering design process to build, test, and improve a prototype;
3. create motion using cams, cranks, linkages, gears, pulleys, or levers;
4. construct and troubleshoot basic breadboard and Arduino circuits;
5. write introductory Arduino programs using inputs, outputs, timing, conditionals, and sensor data;
6. safely connect sensors, actuators, motor drivers, and power sources; and
7. integrate mechanical, electrical, and programming subsystems into a functioning Arduino-based project and explain its design.

## Course Materials

- Course readings, laboratory instructions, and project requirements posted on the course website.
- Arduino Uno-compatible starter kit and a laptop with the Arduino IDE.
- Mechanical-automaton and final-project materials approved by the instructor.
- An engineering notebook for sketches, diagrams, code notes, measurements, and test results.

No textbook purchase is required unless announced by the instructor. Students should not purchase additional project components without instructor approval.

## Course Organization and Expectations

The course meets four times per week for 75 minutes. Lectures introduce the electronics, programming, sensing, actuation, and control concepts needed for the Arduino laboratories. Every numbered laboratory uses Arduino or directly supports an Arduino-controlled system.

- **Project 1 – Mechanical Automata:** A separate, self-directed project completed outside the numbered laboratories. Students independently study a simple mechanism and build a non-motorized prototype that produces repeated animal-inspired motion. The instructor provides the project requirements, examples, checkpoints, and consultation.
- **Final Project – Controller-Based Arduino Project:** Build any approved Arduino-controlled system with a power source, at least one sensor or user input, at least one output or actuator, and repeatable programmed behavior. The project may be a robot, automated device, interactive system, or another approved mechatronic design.

There are no separate homework assignments. Practice and documentation are included in laboratories and projects. Students are expected to attend regularly, work safely, contribute to their team, and keep the engineering notebook current.

## Assessment and Grading

| Assessment category | Weight |
|---|---|
| **Laboratory Activities and Engineering Notebook** | 30% |
| **Project 1: Mechanical Automata** | 30% |
| **Final Project: Controller-Based Arduino Project** | 30% |
| **Peer Evaluation** | **10%** |
| **Total** | **100%** |

| **A** | **B** | **C** | **D** | **F** |
|---|---|---|---|---|
| 90–100% | 80–89.9% | 70–79.9% | 60–69.9% | below 60% |

### Laboratories and Engineering Notebook

Grades reflect preparation, safe participation, completion, troubleshooting, and documentation. Each student must record dated sketches, diagrams, wiring, code notes, measurements, test results, and design changes.

### Project 1: Mechanical Automata

The automaton must use no electric motor or electronic controller, produce repeated animal-inspired motion, and include at least one identifiable mechanism. Grading emphasizes function, design, testing, improvement, documentation, and demonstration.

### Final Project: Controller-Based Arduino Project

The final project must use an Arduino or compatible controller and include a power source, at least one sensor or user input, at least one output or actuator, and programmed decision logic. The project may be a robot, automated mechanism, interactive device, or another approved mechatronic system. Grading emphasizes safe construction, correct operation, integration, testing, documentation, reliability, and presentation.

### Peer Evaluation

Students confidentially evaluate teammates based on preparation, participation, communication, reliability, respect, safety, and contribution. Peer evaluations will inform the grade, but the instructor will review all evaluations and determine the final peer-evaluation grade.

## Laboratory Safety and Equipment Care

Follow all laboratory instructions and use tools and components only for their intended purpose. Wear safety glasses when instructed, secure long hair and loose clothing, disconnect power before changing wiring, and never power motors or other high-current devices directly from Arduino input/output pins. Return equipment in organized condition.

## Course Policies

### Attendance

Regular attendance is expected. In a seven-week course, one absence represents substantial material and project time. Notify the instructor as early as possible and remain responsible for missed work.

### Late and Make-Up Work

Unless stated otherwise, late work loses 10 percentage points per calendar day. Contact the instructor before the deadline when illness or an emergency prevents timely completion. Missed laboratory work may require an alternative assignment.

### Teamwork and Individual Responsibility

Students may collaborate on team projects, but each student must understand the team's design, wiring, code, and test results. Individual notebook entries and reflections must be the student's own work.

### Communication and Conduct

Use Randolph College email and include ENGR 1105 in the subject line. Contribute to a respectful, inclusive, and productive learning environment. Phones and unrelated devices should be put away; laptops may be used for course work.

## Randolph College Policies and Resources

### Accommodations

Students seeking accommodations should contact Larvail Jones, Coordinator of Access Services, at ljones@randolphcollege.edu or 434-947-8132, then meet privately with the instructor after receiving an accommodation letter.

### Title IX and Sexual Misconduct

Faculty members are mandatory reporters. Reports involving sexual misconduct or a campus crime must be shared with the Title IX Coordinator, Jaclyn Beard, at jbeard@randolphcollege.edu or 434-947-8031. The Counseling Center is a confidential resource at counselingcenter@randolphcollege.edu or 434-947-8130.

### Honor Code

Students must submit their own work, identify collaborators, cite sources and software, and never invent measurements or test results. Violations will be handled under the Randolph College Honor Code. See `https://www.randolphcollege.edu/studenthandbook/`.

### Artificial Intelligence

AI tools may be used for brainstorming, grammar, conceptual explanations, or debugging questions. They may not generate submitted designs, code, diagrams, reports, notebook entries, or reflections unless explicitly authorized. Permitted use must be disclosed, and students remain responsible for accuracy, safety, and originality.

## Tentative Course Schedule

The schedule may change based on class progress, component availability, College announcements, or weather. The course website will contain the current version. Project 1 is completed separately from the numbered Arduino laboratories.

| Week | Date | Topic or activity |
|---|---|---|
| **1** | Mon., Oct. 26 | Introduction to robotics and mechatronics; sensors, controllers, actuators, and the sense–decide–act model. Project 1 assigned. |
| **1** | Tue., Oct. 27 | Electrical foundations: voltage, current, resistance, polarity, power, circuit diagrams, and breadboards. |
| **1** | Thu., Oct. 29 | **Lab 1 – Arduino Setup and Blink:** Arduino IDE, board connection, digital output, LEDs, and resistors. |
| **1** | Fri., Oct. 30 | **Lab 2 – Buttons and Digital Input:** Push buttons, pull-up resistors, digital input, and simple decisions. Project 1 concept due. |
| **2** | Mon., Nov. 2 | Arduino program structure: `setup()`, `loop()`, variables, constants, functions, and readable code. |
| **2** | Tue., Nov. 3 | Conditionals, loops, timing with `millis()`, Boolean logic, and Serial Monitor debugging. |
| **2** | Thu., Nov. 5 | **Lab 3 – Digital Control:** Multiple LEDs, button-controlled behavior, timing, and a simple traffic-light sequence. |
| **2** | Fri., Nov. 6 | **Lab 4 – Analog Input and PWM:** Potentiometer input, analog measurements, PWM, and variable LED brightness. |
| **3** | Mon., Nov. 9 | Sensors and measurement: analog versus digital signals, range, resolution, calibration, noise, and thresholds. |
| **3** | Tue., Nov. 10 | Sensor-data processing: Serial Monitor, Serial Plotter, averaging, mapping, and threshold-based decisions. |
| **3** | Thu., Nov. 12 | **Lab 5 – Analog Sensors:** Read and calibrate a light sensor or potentiometer and create an automatic response. |

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| Week | Date | Topic or activity |
|---|---|---|
| **3** | Fri., Nov. 13 | **Lab 6 – Distance Sensing:** Use an ultrasonic sensor to measure distance and build an obstacle-warning system. Project 1 progress check. |
| **4** | Mon., Nov. 16 | Actuators: DC motors, servo motors, and stepper motors; torque, speed, position, current, and actuator selection. |
| **4** | Tue., Nov. 17 | Servo and stepper control; command signals, motion limits, position control, and programmed movement. |
| **4** | Thu., Nov. 19 | **Lab 7 – Servo Control:** Position commands, potentiometer control, and sensor-to-motion behavior. |
| **4** | Fri., Nov. 20 | **Lab 8 – Stepper Motor Control:** Direction, step count, speed, and programmed motion sequences. Project 1 demonstration or video due. |
| **5** | Mon., Nov. 23 | DC motors, transistors, motor drivers, H-bridges, batteries, external power, grounding, and protection. |
| **5** | Tue., Nov. 24 | **Lab 9 – DC Motor and Motor Driver:** Control motor direction and speed safely using Arduino and PWM. |
| **5** | Thu., Nov. 26 | **No class – Thanksgiving break.** |
| **5** | Fri., Nov. 27 | **No class – Thanksgiving break.** |
| **6** | Mon., Nov. 30 | Robot-system design: requirements, block diagrams, subsystem interfaces, power, structure, sensors, and actuators. |
| **6** | Tue., Dec. 1 | Robot behavior: open-loop and closed-loop control, feedback, error, thresholds, and state-based logic. |
| **6** | Thu., Dec. 3 | **Lab 10 – Sensor-Actuator Integration:** Combine an input or sensor with an actuator and test subsystem behavior. |
| **6** | Fri., Dec. 4 | **Lab 11 – Robot Programming:** Organize code into functions and states; begin final-project integration. |
| **7** | Mon., Dec. 7 | System integration, feedback, reliability, safety, and mechanical–electrical–software interactions. |
| **7** | Tue., Dec. 8 | Testing and troubleshooting: fault isolation, test plans, documentation, code comments, and presentation preparation. |
| **7** | Thu., Dec. 10 | **Lab 12 – Final Arduino Project Integration:** Integrate, test, troubleshoot, and improve the controller-based Arduino project. |
| **7** | Fri., Dec. 11 | **Final project demonstrations and team presentations.** |
| **7** | Mon., Dec. 14 | Reflection, peer evaluation, notebook submission, and course synthesis. |

*The instructor may revise this syllabus when necessary. Substantive changes will be announced in class and posted on the course website.*